ME 464 / ME 0460 Kinematics

Complete Formula Sheet

MLS (Murray–Li–Sastry) Product of Exponentials Convention

Paul Martin (PCM) – Verified from Course Notes & Lecture Slides

1 Notation and Conventions

- Twist convention: $\boldsymbol{\xi} = \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{bmatrix}$ (*linear velocity / v on top, angular / ω on bottom*)
- Wrench convention: $\mathbf{F} = \begin{bmatrix} \mathbf{f} \\ \boldsymbol{\tau} \end{bmatrix}$ (*force on top, torque on bottom*)
- Shorthand: $c_i = \cos\theta_i$, $s_i = \sin\theta_i$, $c_{ij} = \cos(\theta_i + \theta_j)$, etc.
- $\|\boldsymbol{\omega}\| = 1$ assumed unless stated otherwise.

2 Linear Algebra Quick Reference

2.1 Transpose and Inverse

$$\mathbf{M}^T = (\text{rows become columns}) \qquad \mathbf{R}^T = \mathbf{R}^{-1} \quad \forall\, \mathbf{R} \in SO(3)$$

$$\mathbf{g}_{ab}^{-1} = \begin{bmatrix} \mathbf{R}_{ab}^T & -\mathbf{R}_{ab}^T \mathbf{p}_{ab} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

2.2 Skew-Symmetric Matrix (Hat Operator $\wedge$)

Maps $\mathbb{R}^3 \to \mathfrak{so}(3)$:

$$\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}^\wedge = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}, \qquad \hat{\boldsymbol{\omega}}\,\mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}, \qquad (\hat{\boldsymbol{\omega}})^T = -\hat{\boldsymbol{\omega}}$$

Vee operator (inverse hat):

$$\boldsymbol{\omega} = \text{vee}(\hat{\boldsymbol{\omega}}) = \begin{bmatrix} W_{32} \\ W_{13} \\ W_{21} \end{bmatrix}$$

Useful identities ($\|\boldsymbol{\omega}\| = 1$):

$$\hat{\boldsymbol{\omega}}^2 = \boldsymbol{\omega}\boldsymbol{\omega}^T - \mathbf{I}, \qquad \hat{\boldsymbol{\omega}}^3 = -\hat{\boldsymbol{\omega}}, \qquad \hat{\boldsymbol{\omega}}^4 = -\hat{\boldsymbol{\omega}}^2$$

3 Rotation Matrices: $SO(3)$

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3\times 3} \mid \mathbf{R}\mathbf{R}^T = \mathbf{I},\ \det(\mathbf{R}) = +1\}$$

Property	**Formula**
Orthogonality	$\mathbf{R}\mathbf{R}^T = \mathbf{I}, \quad \mathbf{R}^{-1} = \mathbf{R}^T$
Determinant	$\det(\mathbf{R}) = +1$
Closure	$\mathbf{R}_1\mathbf{R}_2 \in SO(3)$
Composition	$\mathbf{R}_{ac} = \mathbf{R}_{ab}\mathbf{R}_{bc}$
Vector transform	$\mathbf{v}_a = \mathbf{R}_{ab}\mathbf{v}_b$

3.1 Rodrigues' Formula (Exponential Map for Rotations)

For $\boldsymbol{\omega} \in \mathbb{R}^3$ with $\|\boldsymbol{\omega}\| = 1$, rotation by angle θ:

$$\boxed{e^{\hat{\boldsymbol{\omega}}\theta} = \mathbf{I} + \sin\theta\,\hat{\boldsymbol{\omega}} + (1 - \cos\theta)\,\hat{\boldsymbol{\omega}}^2}$$

3.2 Inverse Map (Extract θ and $\boldsymbol{\omega}$ from R)

$$\theta = \arccos\left(\frac{\operatorname{trace}(\mathbf{R}) - 1}{2}\right), \qquad \boldsymbol{\omega} = \frac{1}{2\sin\theta}\operatorname{vee}(\mathbf{R} - \mathbf{R}^T) \quad (\theta \neq 0, \pi)$$

3.3 Principal Rotation Matrices

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}, \quad R_y(\theta) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}, \quad R_z(\theta) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4 Homogeneous Transformations: $SE(3)$

$$SE(3) = \{(\mathbf{R}, \mathbf{p}) \mid \mathbf{R} \in SO(3),\ \mathbf{p} \in \mathbb{R}^3\}$$

$$\mathbf{g}_{ab} = \begin{bmatrix} \mathbf{R}_{ab} & \mathbf{p}_{ab} \\ \mathbf{0}^T & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4}, \qquad \mathbf{g}_{ab}^{-1} = \begin{bmatrix} \mathbf{R}_{ab}^T & -\mathbf{R}_{ab}^T\mathbf{p}_{ab} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Point transformation (using homogeneous coordinates):

$$\begin{bmatrix} \mathbf{p}_a \\ 1 \end{bmatrix} = \mathbf{g}_{ab}\begin{bmatrix} \mathbf{p}_b \\ 1 \end{bmatrix}, \qquad \mathbf{g}_{ac} = \mathbf{g}_{ab}\mathbf{g}_{bc}$$

5 Twists and Screw Motions

5.1 Twist Vector $\boldsymbol{\xi} \in \mathbb{R}^6$

Revolute joint (rotation about axis $\boldsymbol{\omega}$ through point $\mathbf{q}$):

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{bmatrix}, \qquad \mathbf{v} = -\boldsymbol{\omega} \times \mathbf{q} = -\hat{\boldsymbol{\omega}}\,\mathbf{q}$$

Prismatic joint (translation along unit direction $\mathbf{v}$):

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{v} \\ \mathbf{0} \end{bmatrix}, \qquad \|\mathbf{v}\| = 1$$

5.2 Twist Hat Map: $\boldsymbol{\xi}^\wedge \in \mathbb{R}^{4\times 4}$

$$\boldsymbol{\xi}^\wedge = \begin{bmatrix} \hat{\boldsymbol{\omega}} & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

5.3 Exponential Map for Twists

Revolute ($\|\boldsymbol{\omega}\| = 1$):

$$\boxed{e^{\hat{\boldsymbol{\xi}}\,\theta} = \begin{bmatrix} e^{\hat{\boldsymbol{\omega}}\theta} & \mathbf{p}(\theta) \\ \mathbf{0}^T & 1 \end{bmatrix}}$$

where

$$\mathbf{p}(\theta) = \left(\mathbf{I} - e^{\hat{\boldsymbol{\omega}}\theta}\right)(\boldsymbol{\omega} \times \mathbf{v}) + \boldsymbol{\omega}\,(\boldsymbol{\omega}^T\mathbf{v})\,\theta$$

When $\boldsymbol{\omega}^T\mathbf{v} = 0$ (pure rotation, zero pitch):

$$\mathbf{p}(\theta) = \left(\mathbf{I} - e^{\hat{\boldsymbol{\omega}}\theta}\right)(\boldsymbol{\omega} \times \mathbf{v})$$

Prismatic ($\boldsymbol{\omega} = \mathbf{0}$, displacement d):

$$e^{\hat{\boldsymbol{\xi}}\,d} = \begin{bmatrix} \mathbf{I} & \mathbf{v}\,d \\ \mathbf{0}^T & 1 \end{bmatrix}$$

6 Forward Kinematics: Product of Exponentials (POE)

Tool frame transform for an n-DOF manipulator:

$$\boxed{\mathbf{g}_{st}(\boldsymbol{\theta}) = e^{\hat{\boldsymbol{\xi}}_1\theta_1}\, e^{\hat{\boldsymbol{\xi}}_2\theta_2}\, \cdots\, e^{\hat{\boldsymbol{\xi}}_n\theta_n}\, \mathbf{g}_{st}(0)}$$

- $\boldsymbol{\xi}_i$: twist for joint i evaluated in the **zero configuration** (home position).
- $\mathbf{g}_{st}(0)$: tool frame relative to spatial frame at $\boldsymbol{\theta} = \mathbf{0}$ (measured from robot geometry).
- Multiply **left-to-right**: joint 1 outermost, joint n innermost (before $\mathbf{g}_{st}(0)$).

Procedure to find twists $\boldsymbol{\xi}_i$:

1. Find joint axis direction $\boldsymbol{\omega}_i$ (revolute) or translation direction $\mathbf{v}_i$ (prismatic) in the zero configuration.
2. Find any point $\mathbf{q}_i$ on joint axis i.
3. Compute: $\mathbf{v}_i = -\boldsymbol{\omega}_i \times \mathbf{q}_i$ (revolute) or $\boldsymbol{\omega}_i = \mathbf{0}$ (prismatic).

7 Adjoint Transformation

For $\mathbf{g} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3)$:

$$\boxed{\mathrm{Ad}_{\mathbf{g}} = \begin{bmatrix} \mathbf{R} & \hat{\mathbf{p}}\mathbf{R} \\ \mathbf{0} & \mathbf{R} \end{bmatrix} \in \mathbb{R}^{6\times 6}}$$

Inverse:

$$\mathrm{Ad}_{\mathbf{g}}^{-1} = \mathrm{Ad}_{\mathbf{g}^{-1}} = \begin{bmatrix} \mathbf{R}^T & -\mathbf{R}^T\hat{\mathbf{p}} \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix}$$

Verification: $\mathrm{Ad}_{\mathbf{g}} \cdot \mathrm{Ad}_{\mathbf{g}}^{-1} = \mathbf{I}_{6\times 6}$. *Note* $\hat{\mathbf{p}}$ *is the skew-symmetric matrix of* $\mathbf{p}$.

Velocity mapping:

$$\mathcal{V}^a = \mathrm{Ad}_{\mathbf{g}_{ab}}\, \mathcal{V}^b, \qquad \mathcal{V}^b = \mathrm{Ad}_{\mathbf{g}_{ab}}^{-1}\, \mathcal{V}^a$$

8 Spatial and Body Jacobians

8.1 Spatial Jacobian $\mathbf{J}_{st}^s(\boldsymbol{\theta})$

Relates joint velocities $\dot{\boldsymbol{\theta}}$ to the **spatial velocity** of the tool:

$$\mathcal{V}^s = \mathbf{J}_{st}^s(\boldsymbol{\theta})\, \dot{\boldsymbol{\theta}}$$

$$\boxed{\mathbf{J}_{st}^s(\boldsymbol{\theta}) = \left[\, \boldsymbol{\xi}_1 \;\middle|\; \mathrm{Ad}_{G_1}\boldsymbol{\xi}_2 \;\middle|\; \mathrm{Ad}_{G_2}\boldsymbol{\xi}_3 \;\middle|\; \cdots \;\middle|\; \mathrm{Ad}_{G_{n-1}}\boldsymbol{\xi}_n \,\right]}$$

where

$$G_i = e^{\hat{\boldsymbol{\xi}}_1\theta_1} \cdots e^{\hat{\boldsymbol{\xi}}_i\theta_i} \qquad (G_0 = \mathbf{I})$$

Column-by-column: $\mathbf{J}_i^s = \mathrm{Ad}_{G_{i-1}}\, \boldsymbol{\xi}_i$

Note: only the joints before *joint i contribute to G_{i-1}. Column 1 is always $\boldsymbol{\xi}_1$ (no preceding joints).*

8.2 Body Jacobian $\mathbf{J}_{st}^b(\boldsymbol{\theta})$

Relates joint velocities to the **body velocity** of the tool:

$$\mathcal{V}^b = \mathbf{J}_{st}^b(\boldsymbol{\theta})\, \dot{\boldsymbol{\theta}}$$

$$\boxed{\mathbf{J}_{st}^b(\boldsymbol{\theta}) = \mathrm{Ad}_{\mathbf{g}_{st}(\boldsymbol{\theta})^{-1}}\, \mathbf{J}_{st}^s(\boldsymbol{\theta})}$$

8.3 Velocity Summary

Velocity	**Frame**	**Formula**
Spatial	Spatial frame (fixed)	$\mathcal{V}^s = \mathbf{J}^s\, \dot{\boldsymbol{\theta}}$
Body	Tool frame (moving)	$\mathcal{V}^b = \mathbf{J}^b\, \dot{\boldsymbol{\theta}}$
Conversion	$\mathcal{V}^s \leftrightarrow \mathcal{V}^b$	$\mathcal{V}^b = \mathrm{Ad}_{\mathbf{g}_{st}^{-1}}\, \mathcal{V}^s$

9 Wrenches

Wrench convention: $\mathbf{F} = \begin{bmatrix} \mathbf{f} \\ \boldsymbol{\tau} \end{bmatrix}$ (force on top, torque on bottom).

Power invariance (same in any frame):

$$(\mathbf{F}^b)^T \mathcal{V}^b = (\mathbf{F}^a)^T \mathcal{V}^a$$

Wrench transformation (from spatial frame a to body frame b):

$$\mathbf{F}^b = \mathrm{Ad}_{\mathbf{g}_{ab}}^T \mathbf{F}^a$$

Spatial wrench at origin due to force $\mathbf{f}$ applied at point $\mathbf{r}$:

$$\mathbf{F}^a = \begin{bmatrix} \mathbf{f} \\ \mathbf{r} \times \mathbf{f} \end{bmatrix}$$

Jacobian-wrench relationship (static equilibrium):

$$\boldsymbol{\tau} = (\mathbf{J}^s)^T \mathbf{F}^s = (\mathbf{J}^b)^T \mathbf{F}^b$$

where $\boldsymbol{\tau}$ is the vector of joint torques/forces.

10 Working Through a Problem: Step-by-Step Checklist

1. **Draw** the robot in its zero configuration. Label all link lengths and frame axes.
2. **Find** $\mathbf{g}_{st}(0)$: read the tool position and orientation from the geometry at home.
3. **Find twists** $\boldsymbol{\xi}_i$: for each joint, identify $\boldsymbol{\omega}_i$ (or $\mathbf{v}_i$) and a point $\mathbf{q}_i$ on its axis in the zero config.
4. **Compute** $\mathbf{g}_{st}(\boldsymbol{\theta})$: multiply POE chain and simplify with MATLAB (`simplify`).
5. **Compute** $\mathbf{J}^s$: build column by column using $\mathrm{Ad}_{G_{i-1}} \boldsymbol{\xi}_i$.
6. **Compute** $\mathbf{J}^b$ (if needed): $\mathbf{J}^b = \mathrm{Ad}_{\mathbf{g}_{st}^{-1}} \mathbf{J}^s$.

11 Key MATLAB Functions (Course Sub-Functions)

Function	Purpose / Output
`expt_PCM(xi, theta)`	$e^{\hat{\boldsymbol{\xi}}\theta}$ – 4×4 twist exponential
`expr_PCM(omega, theta)`	$e^{\hat{\boldsymbol{\omega}}\theta}$ – 3×3 rotation matrix
`wedge_PCM(xi)`	$\hat{\boldsymbol{\xi}}$ – 4×4 twist hat map
`Ad(g)`	$\mathrm{Ad}_{\mathbf{g}}$ – 6×6 adjoint matrix
`matlabFunction(g, 'File', 'fname', 'Vars', {[th1,...,thn]})`	Generates numeric FK function

12 Reference: HW4 Robot (2-DOF, Revolute–Revolute)

Geometry: offset d_0 along z, then d_1 along x, then d_2 radially, tool at d_3.

$$\mathbf{R}(\theta) = \begin{bmatrix} 0 & 1 & 0 \\ -s\theta & 0 & c\theta \\ c\theta & 0 & s\theta \end{bmatrix}, \qquad \hat{\mathbf{p}}(\theta) = \begin{bmatrix} 0 & -d_2 s\theta & d_1 + d_2 c\theta \\ d_2 s\theta & 0 & -d_0 \\ -(d_1 + d_2 c\theta) & d_0 & 0 \end{bmatrix}$$

$$\mathbf{g}_{ab}^{-1}(\theta) = \begin{bmatrix} 0 & -s\theta & c\theta & d_1 s\theta \\ 1 & 0 & 0 & -d_0 \\ 0 & c\theta & s\theta & -d_1 c\theta - d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Spatial wrench (W = weight, y-gravity in spatial frame):

$$\mathbf{F}^a(\theta) = \begin{bmatrix} 0 \\ -W \\ 0 \\ W(d_2 s\theta + d_3 c\theta) \\ 0 \\ -W d_0 \end{bmatrix}, \qquad \mathbf{F}^b(\theta) = \begin{bmatrix} W s\theta \\ 0 \\ -W c\theta \\ 0 \\ W d_3 c\theta \\ 0 \end{bmatrix}$$

13 Reference: HW5 Stanford Manipulator (6-DOF)

Link dimensions: $d_1 = 40\,\text{cm}$, $d_2 = 50\,\text{cm}$, $d_3 = 60\,\text{cm}$ (prismatic), $d_6 = 30\,\text{cm}$.
Joint 3 is **prismatic** (all others revolute).

$$\mathbf{g}_{st}(0) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 + d_6 \\ 1 & 0 & 0 & d_1 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Twists ($[\mathbf{v}; \boldsymbol{\omega}]$ convention):

$$\boldsymbol{\xi}_1 = [0,\ 0,\ 0;\ 0,\ 0,\ 1]^T \qquad \boldsymbol{\xi}_2 = [0,\ d_1,\ 0;\ 1,\ 0,\ 0]^T$$
$$\boldsymbol{\xi}_3 = [0,\ 0,\ 1;\ 0,\ 0,\ 0]^T \text{ (prismatic)} \qquad \boldsymbol{\xi}_4 = [d_2,\ 0,\ 0;\ 0,\ 0,\ 1]^T$$
$$\boldsymbol{\xi}_5 = [0,\ d_1{+}d_3,\ -d_2;\ 1,\ 0,\ 0]^T \qquad \boldsymbol{\xi}_6 = [-(d_1{+}d_3),\ 0,\ 0;\ 0,\ 1,\ 0]^T$$

Verification (MATLAB): `gstthPCM([0 0 0 0 0 0])` → `[0 1 0 0; 0 0 1 80; 1 0 0 100; 0 0 0 1]`

14 Reference: HW6 2-DOF Manipulator (Spatial Jacobian)

Joint axes and twists (z-rotation then y-rotation; $q_2 = [d_1; 0; 0]$):

$$\boldsymbol{\xi}_1 = [0;\ 0;\ 0;\ 0;\ 0;\ 1]^T, \qquad \boldsymbol{\xi}_2 = [0;\ 0;\ d_1;\ 0;\ 1;\ 0]^T$$

Home configuration: $\mathbf{g}_{st}(0) = [\mathbf{I} \mid [d_1{+}d_2;\ d_3;\ 0];\ \mathbf{0}\ 1]$
Spatial Jacobian (column 1 = $\boldsymbol{\xi}_1$; column 2 = $\mathrm{Ad}_{e^{\hat{\boldsymbol{\xi}}_1\theta_1}}\boldsymbol{\xi}_2$):

$$\mathbf{J}^s_{st}(\theta_1, \theta_2) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & d_1 \\ 0 & -s_1 \\ 0 & c_1 \\ 1 & 0 \end{bmatrix}$$

Derivation: $e^{\hat{\boldsymbol{\xi}}_1\theta_1}$ *has* $\mathbf{p} = \mathbf{0}$ *so* $\mathrm{Ad} = \mathrm{blkdiag}(R_z, R_z)$. *Apply to* $\boldsymbol{\xi}_2$*:* $R_z[0;0;d_1] = [0;0;d_1]$ *and* $R_z[0;1;0] = [-s_1; c_1; 0]$.

Verified numerically. All formulas follow MLS (Murray–Li–Sastry) Product of Exponentials convention with $\boldsymbol{\xi} = [\mathbf{v}; \boldsymbol{\omega}]$ *and* $\mathbf{F} = [\mathbf{f}; \boldsymbol{\tau}]$.